Flanker Task EEG functional data hypothesis testing

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Introduction

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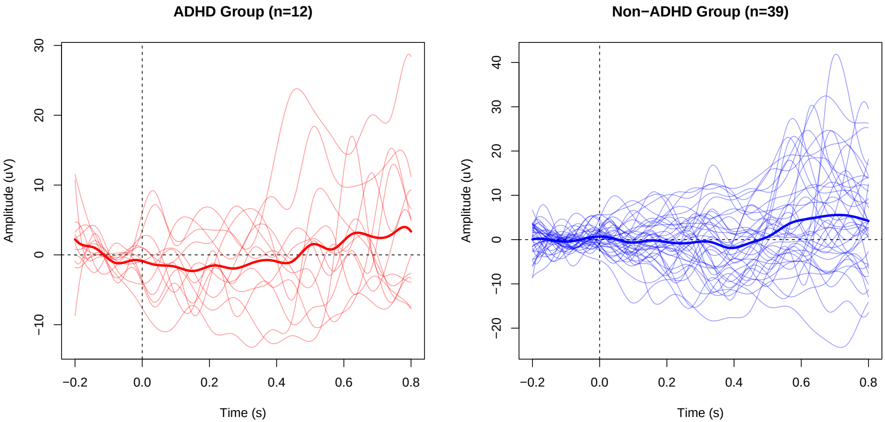
- In this part of the project we test whether the functional EEG response curves during a Flanker task differ significantly between individuals with ADHD symptoms (n=12) and without (n=39).

$$
\begin{aligned}
H_0 &: \mu_{\text{ADHD}}(t) = \mu_{\text{non-ADHD}}(t), \quad \forall\, t \in [-0.2,\, 0.8] \\
H_1 &: \mu_{\text{ADHD}}(t) \neq \mu_{\text{non-ADHD}}(t), \quad \text{for some } t
\end{aligned}
$$

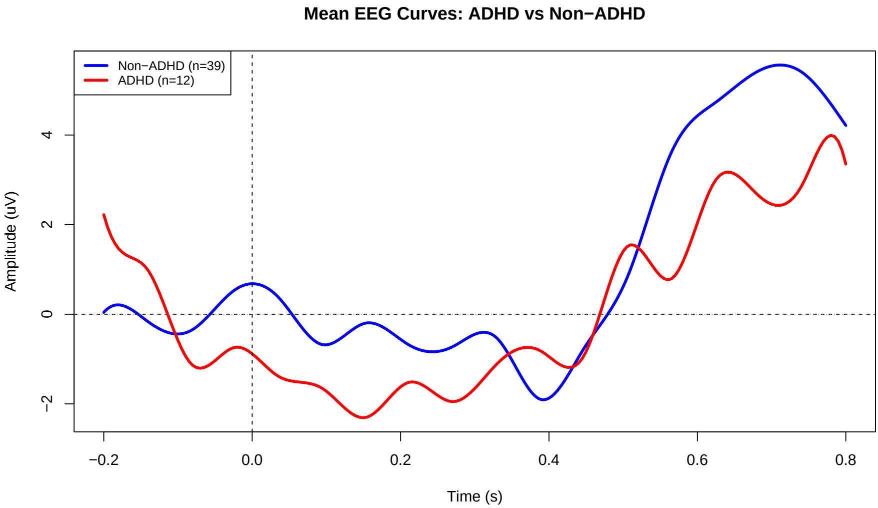
Group Comparison

Group Comparison



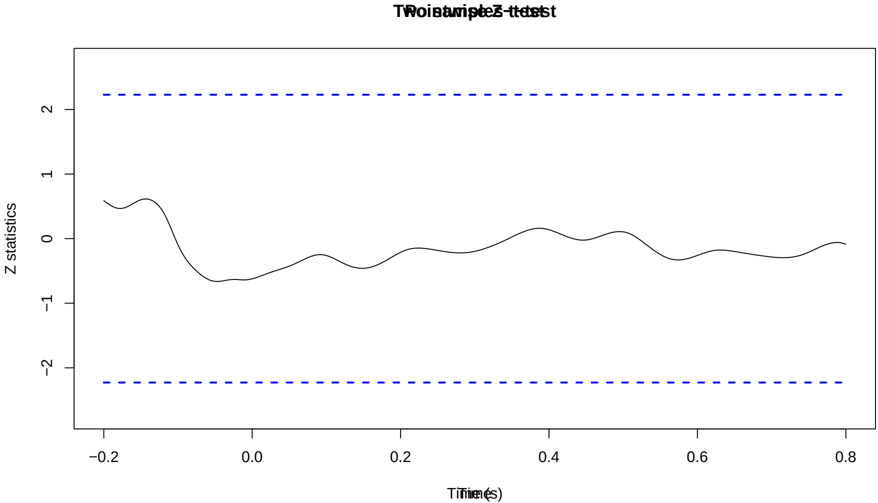
Mean Curves

Mean Curves


Mean EEG Curves: ADHD vs Non−ADHD
Non−ADHD (n=39)
ADHD (n=12)
Amplitude (uV)
4
2
0
−2
−0.2
0.0
0.2
0.4
0.6
0.8
Time (s)

Hypothesis testing

Pointwise Z-test



L2-norm Test

The L2-norm test measures the total integrated squared difference between the two group mean curves:

$$T_{L^2} = \frac{nm}{n+m} \int [\bar{x}(t) - \bar{y}(t)]^2 \, dt$$

Method	**Statistic**	**p-value**
Naive (χ^2 approximation)	10882.89	1.0000
Bootstrap (500 replications)	10882.89	0.7260

Both methods yield large p-values ($p > 0.05$), indicating no significant global difference between the ADHD and non-ADHD mean curves.

F-type Test

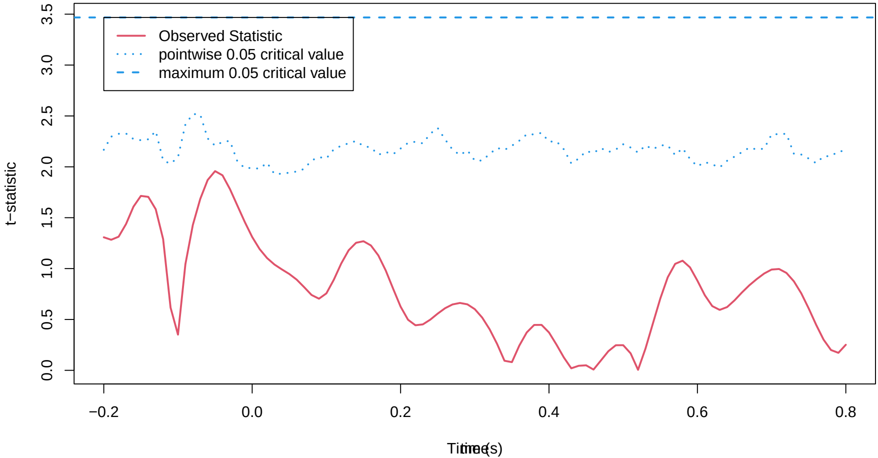
The F-type test normalizes the L2 statistic by the trace of the covariance operator:

$$F = \frac{T_{L^2}}{\mathrm{tr}(\hat{\Sigma})}$$

Method	**Statistic**	**p-value**
Naive (F approximation)	0.0818	1.0000
Bootstrap (500 replications)	0.0818	0.7140

The F-type test accounts for the covariance structure of the data. Large p-values ($p > 0.05$) confirm no significant difference between the two groups.

Permutation Test



Summary

Summary

Test	p-value
Pointwise Z-test (max $\|Z\| = 0.66$)	–
L2-norm test (naive)	1.0000
L2-norm test (bootstrap)	0.7260
F-type test (naive)	1.0000
F-type test (bootstrap)	0.7140
Permutation test	0.5250

Conclusion: All tests fail to reject H_0 at $\alpha = 0.05$. There is no significant difference between the EEG response curves of individuals with and without ADHD symptoms during the Flanker task at channel FC1.

This may be due to the small ADHD group size (n=12), the use of a single EEG channel, or the nature of the ADHD classification.

Thank you for your time!